

MODULAR OPERATION THEATRE



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ABOUT US

- ⚡ Pluridisciplinary organization set-up in 2011 backed by young, enthusiastic, experienced and well-qualified team of Bio-medical Engineers having sound experience in healthcare sector.
- ⚡ 100+ fully satisfied customers in a short span of 7 years spread across most of Gujarat.
- ⚡ Consultant Bio-medical Engineering Services at P.P. Savani Hospital & SDA Diamond Hospital.

VISION & MISSION

VISION :

To be the leading manufacturer, supplier, service provider & technological consultant of Modular O.T., Medical Gas Pipeline System and OT-ICU-NICU-CSSD equipments.

MISSION :

By giving innovative and affordable solutions and products with timely post-sales services enhancing the quality of treatment and services provided by the hospital to the patients.

PRODUCTS & SERVICES

- ⚡ Modular O.T.
- ⚡ Medical Gas Pipeline System
- ⚡ Track & Curtain System

PRIME CLIENT LIST

- 📞 P.P. Savani Multi-speciality Hospital, Surat
- 📞 SDA Diamond Hospital, Surat
- 📞 Tri-star Hospital, Surat
- 📞 Mission Hospital, Surat
- 📞 Smt. Maroliya Hospital, Navsari
- 📞 BAPS Pramukh Swami Hospital, Surat
- 📞 Nutan Medical College & Hospital, Visnagar
- 📞 GCS Medical College & Hospital, Ahmedabad
- 📞 HCG Cancer Hospital, Ahmedabad
- 📞 LG General Hospital, Ahmedabad
- 📞 Vibrant Hospital, Vapi
- 📞 Apex Dialysis Center, Palanpur
- 📞 Sunshine Global Hospital, Vadodara
- 📞 Coral Hospital, Vadodara
- 📞 ASG Eye Hospital, Surat
- 📞 Bhagyoday Hospital, Kadi
- 📞 Janta Trust Hospital, Patan
- 📞 Sanjivani Hospital, Modasa
- 📞 Gita Hospital, Patan
- 📞 Navjivan Hospital, Surat
- 📞 Pramukh Hospital, Surendranagar
- 📞 Shreeji Hospital, Dhangradra
- 📞 Medico Multi-speciality Hospital, Surendranagar

MOT - CONCEPT

A jointless and seamless structure is designed and installed with the aim to provide a clean and hygienic (0.3 micron) environment inside the O.T. to avoid any kind of infection to the patient during the surgery controlling the temperature, pressure and humidity.

It employs control over the particulate matter in air with temperature, humidity and pressure control as required thereby reducing bacteria by a factor of 15,000.

DIVISIONS OF O.T.

- **Superspeciality O.T.** – Neuroscience, Orthopedics, Cardiothoracic & Transplant surgery
- **General O.T.** – Laparoscopy, Ophthalmology, Gynecology, ENT, Dental, etc.

CLASSIFICATION OF CLEAN ROOMS

USA Class	Maximum Particles/ft ³					ISO Equivalent
	≥0.1 μm	≥0.2 μm	≥0.3 μm	≥0.5 μm	≥5 μm	
1	35	7.5	3	1	0.007	ISO 3
10	350	75	30	10	0.07	ISO 4
100	3,500	750	300	100	0.7	ISO 5
1,000	35,000	7,500	3000	1,000	7	ISO 6
10,000	350,000	75,000	30,000	10,000	70	ISO 7
100,000	3.5×10 ⁶	750,000	300,000	100,000	700	ISO 8

NABH SPECIFICATIONS

Features	Superspeciality O.T.	General O.T.
Temperature (deg Celsius)	21 $\pm$ 3	21 $\pm$ 3
Pressure (Pascal)	10-15	10-15
Humidity (Rh)	40-60%	40-60%
Air Filtration (Micron)	10 μ Pre-Filters at AHU; 5 μ Fine-filters at AHU & 0.3 μ HEPA Filter above OT Table	10 μ Pre-Filters at AHU; 5 μ Fine-filters at AHU & 0.3 μ HEPA Filter above OT Table
Plenum Box Size (Feet)	8 x 6	6 x 4
Class US/ISO	100/ISO 5	1000/ISO 6
Air changes per hour	$\geq$ 25	$\geq$ 25
Fresh Air Component	$\geq$ 5	$\geq$ 5
Air Velocity (FPM)	90-120	90-120

NABH GUIDELINES

- ⚡ Dedicated A.H.U. for every OT
- ⚡ No Window & Split AC
- ⚡ Wall, Ceiling & Flooring material should be non-porous, jointless-seamless without 90° corners & washable with anti-bacterial and anti-fungal properties.
- ⚡ Cleaning of pre-filters
- ⚡ Validation

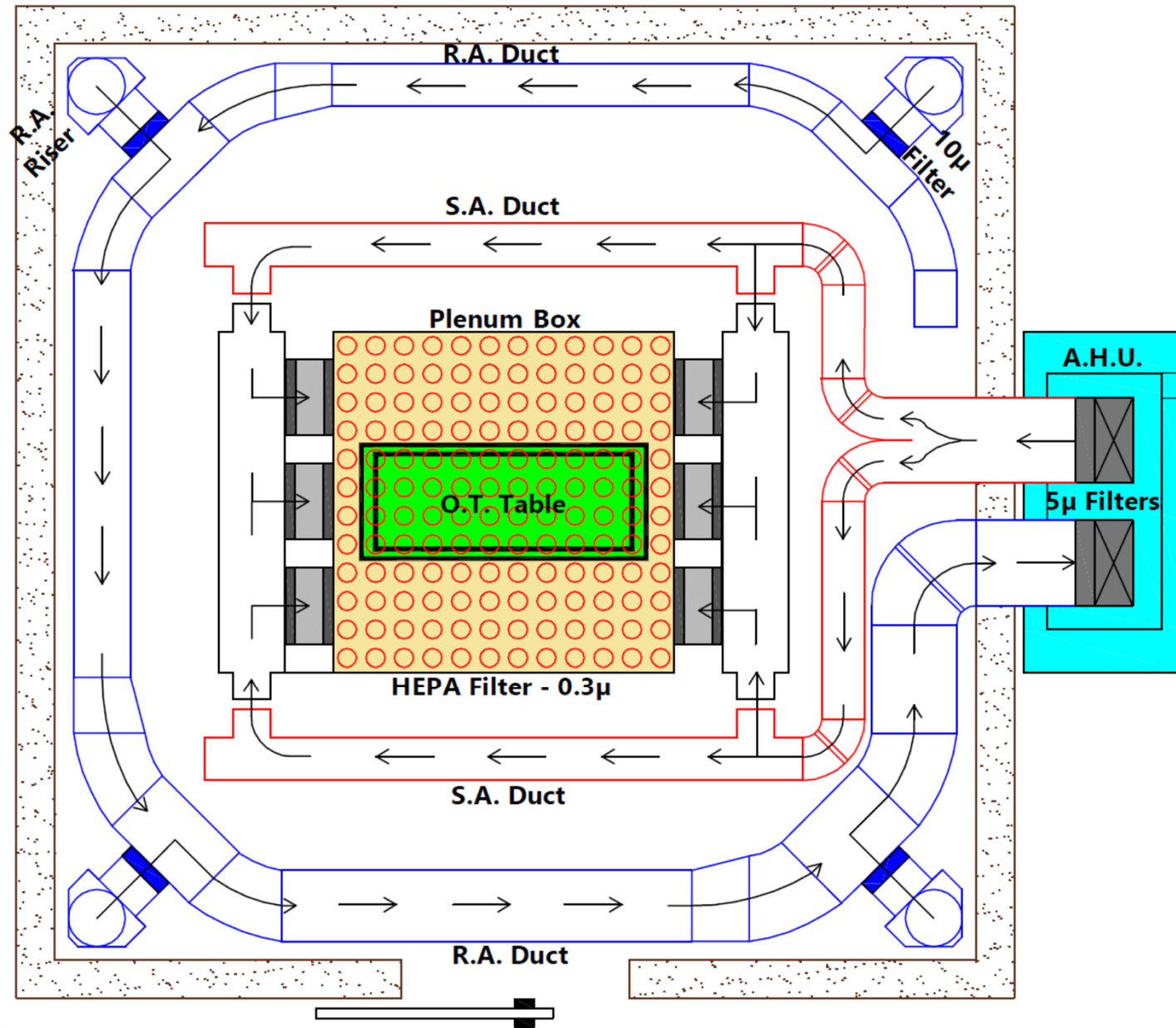
MOT - COMPONENTS

- ⚡ **Primary Component** - Vertical Laminar Air Flow kills any bacteria present inside the O.T. area giving 0.3 micron clean air.
- ⚡ **Secondary Component** - Complete jointless and seamless structure without any 90° corners for the prevention of bacterial formation in the corners, joints or cracks.

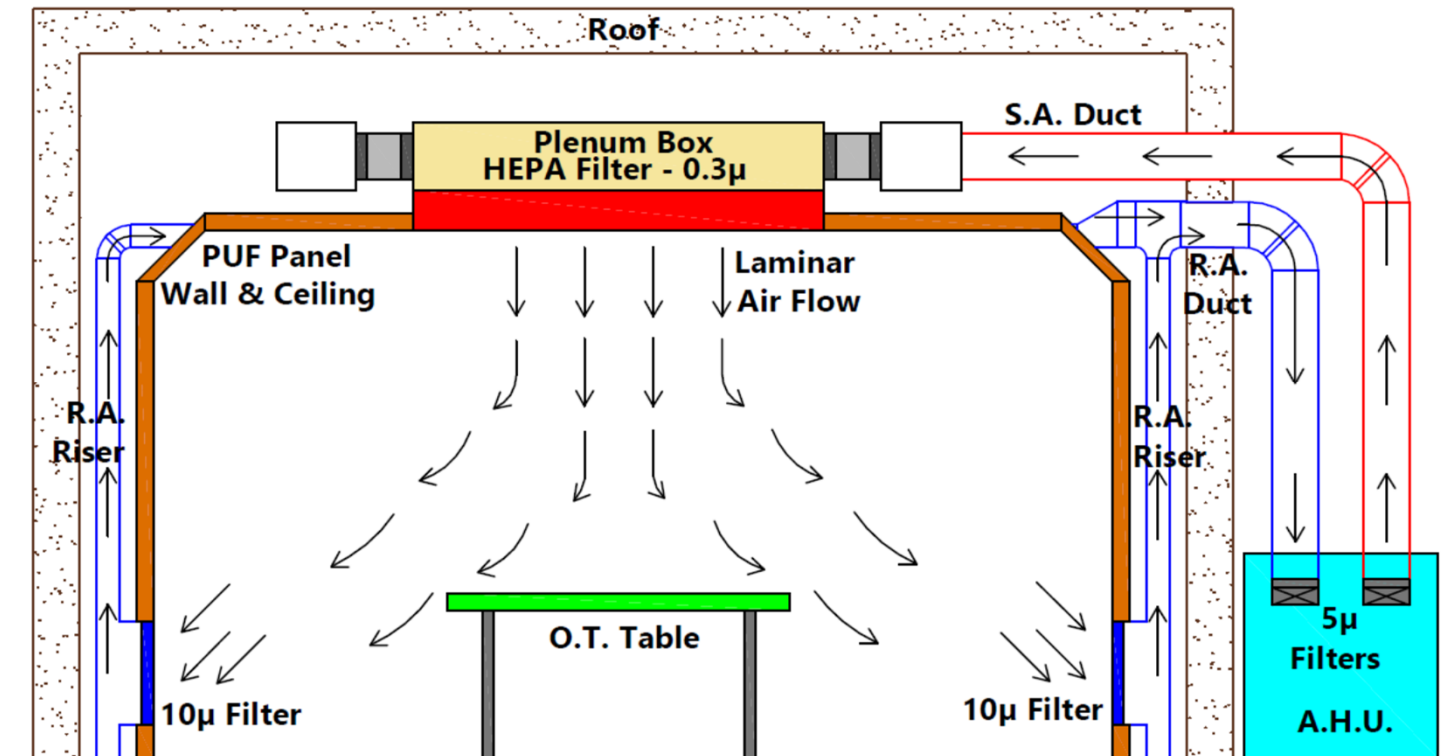
LAMINAR AIR FLOW (LAF)

- ⚡ Air Handling Unit (A.H.U.)
- ⚡ AC connected to AHU
- ⚡ HEPA Filters
- ⚡ Plenum(Filter Housing) Box above OT Table
- ⚡ Ducting

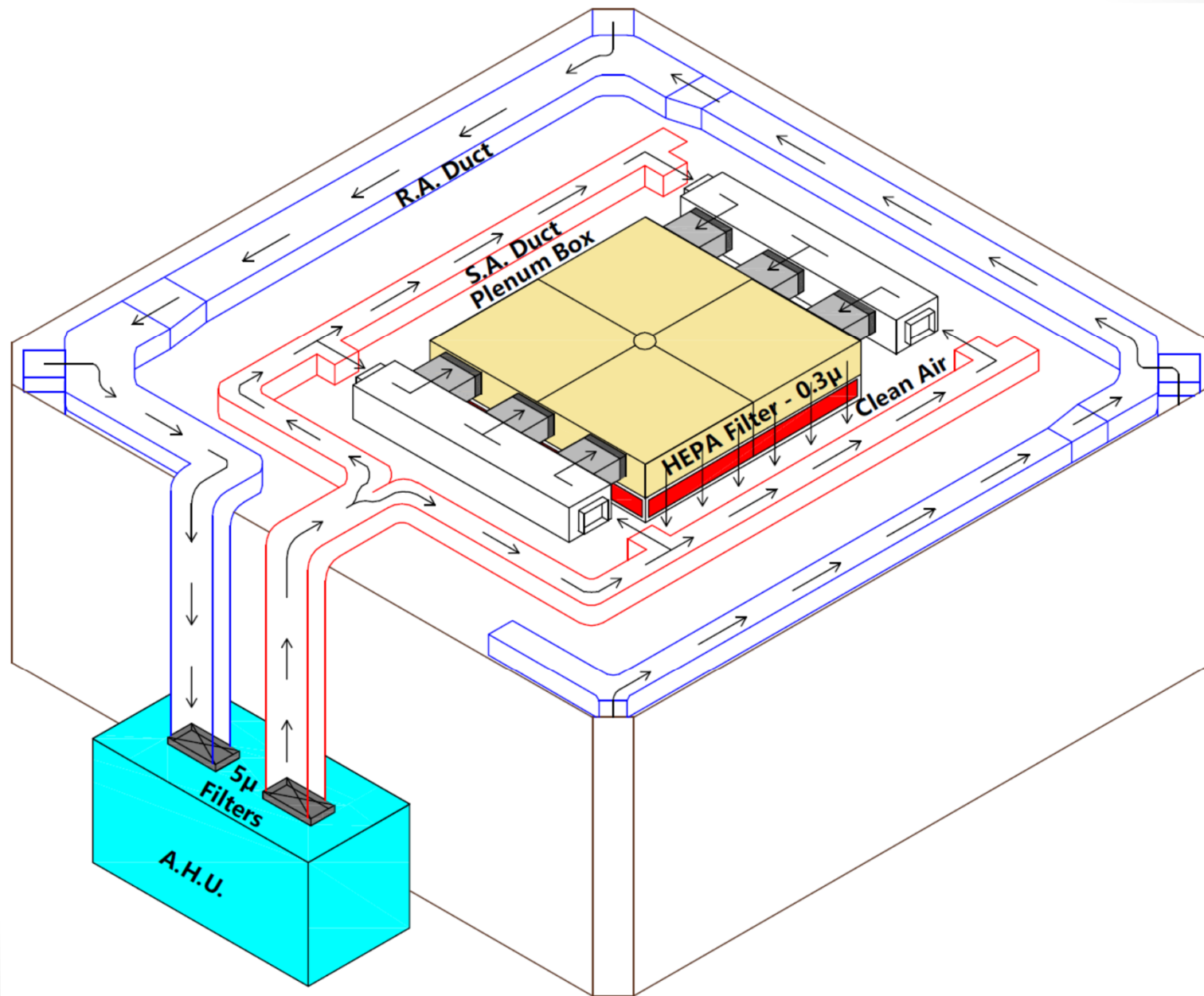
LAF DESIGN - PLAN



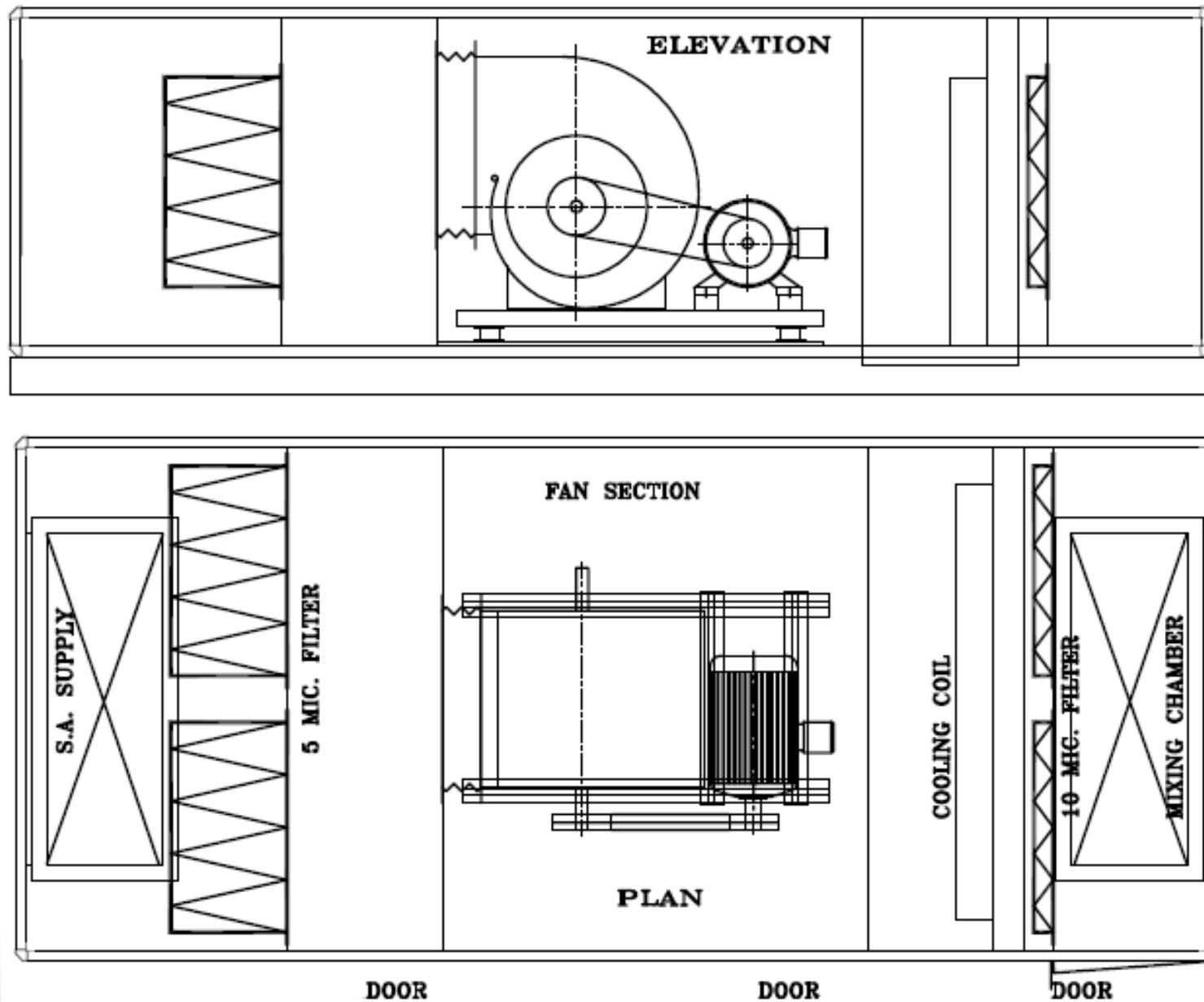
LAF DESIGN - ELEVATION



LAF DESIGN - ISOMETRIC VIEW



A.H.U. DESIGN



A.H.U. & PLENUM BOX



A.H.U.



Diffuser Cloth & SS Grill Plenum Box

JOINTLESS & SEAMLESS STRUCTURE

WALL MATERIAL

- ⚡ Corian Panels
- ⚡ Sandwich PUF Panels (GI, ACP, HPL, SS)
- ⚡ ACP/HPL Sheet
- ⚡ Vinyl Wall
- ⚡ Anti-bacterial Paint & Epoxy

CEILING MATERIAL

- ⚡ Corian Panels
- ⚡ Sandwich PUF Panels (GI-GI, HPL-GI, HPL-HPL, SS-GI, SS-SS)
- ⚡ ACP/HPL Sheet

FLOORING MATERIAL

- ⚡ Vinyl
- ⚡ Epoxy

ANTI-BACTERIAL PAINT

SPECIFICATIONS

Two-Three layers of paint with the D.F.T. of 2-3 micron with one layer of primer. Joints & corners are sealed and rounded-off using putty.

TYPES

- ⚡ Hand Paint (Roller)
- ⚡ Machine Paint (Spray)

MERITS

- ⚡ Cost-effective
- ⚡ Glare Free

DEMERITS

- ⚡ Life : Hand Paint (4-5 years); Machine Paint (7-8 years)
- ⚡ Moisture & Abrasion Resistance – Very Low

EPOXY WALL

MERITS

- ⚡ Resistant to moisture and abrasion
- ⚡ Cost-effective

DEMERITS

- ⚡ Life of 4-5 years
- ⚡ Glare
- ⚡ Not anti-static
- ⚡ Chances of crack
- ⚡ ESD Property not available

VINYL WALLS

SPECIFICATIONS

Completely smooth and undulated surface is must.

MERITS

- ⚡ Cost-effective
- ⚡ Life of 10-12 years
- ⚡ Can be re-cycled
- ⚡ Glare Free

DEMERITS

- ⚡ Chances of air bubbles
- ⚡ Moisture & Abrasion Resistant - Medium

ACP/HPL SHEET

SPECIFICATIONS

ACP/HPL Sheet of 3/4 mm thickness are erected on aluminum framing. Joints & corners are in radius & sealed using Silicon.

MERITS

- ⚡ Cost-effective
- ⚡ Resistant to moisture and abrasion
- ⚡ Life of 10-12 years
- ⚡ Glare Free

DEMERITS

- ⚡ Silicon wears out on longer run
- ⚡ No insulation backing
- ⚡ Chances of crack

GI PUF PANELS

TYPES

- ⚡ Pre-coated (which is not anti-bacterial)
- ⚡ Hand/Machine Paint on site.

SPECIFICATIONS

GI Sheets are sandwiched between a layer of PUF (Polyurethane foam) of thickness 30/40mm. Joints & corners are in radius & sealed using silicon or paint is applied.

MERITS

- ⚡ Resistant to moisture and abrasion
- ⚡ Life of 12-15 years (Paint Life as actual)
- ⚡ Glare Free

DEMERITS

- ⚡ Costly
- ⚡ Chances of rusting
- ⚡ Silicon wears out on longer run

ACP/HPL PUF PANELS

SPECIFICATIONS

ACP/HPL Sheets are sandwiched between a layer of PUF (Poly-urethane foam) of thickness 30/40mm. Joints & corners are in radius & sealed using silicon.

MERITS

- ⚡ Resistant to moisture and abrasion
- ⚡ Life of 12-15 years
- ⚡ Glare Free

DEMERITS

- ⚡ Costly
- ⚡ Silicon wears out on longer run

SS PUF PANELS

SPECIFICATIONS

SS Panels are sandwiched between a layer of PUF (Polyurethane foam) of thickness 30/40mm. Joints & corners are in radius & sealed using Silicon.

MERITS

- ⚡ Resistant to moisture and abrasion
- ⚡ Life of 12-15 years

DEMERITS

- ⚡ Costly
- ⚡ Glare
- ⚡ Silicon wears out on longer run

CORIAN PANELS

SPECIFICATIONS

Corian Panels of 6/12mm thickness are erected on ply attached to aluminum framing. Joints & corners are in radius & sealed using PTFE chemicals.

MERITS

- ⚡ Resistant to moisture and abrasion
- ⚡ Life of 20-25 years
- ⚡ Glare Free
- ⚡ Complete Jointless & Seamless surface without any 90° corners

DEMERITS

- ⚡ Costly

MATERIAL COMPARISON

Features	Paint	Epoxy Wall	Vinyl Walls	ACP/HPL Sheets	GI PUF Panels	ACP/HPL Puf Panel	SS PUF Panels	Corian Panels
Anti-Bacterial	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Anti-Fungal	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Jointless & Seamless	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Non-Porous	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Washable	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Thickness	2-3μ	65-70μ	1mm	3/4mm	40mm	44mm	40mm	6/12mm
Life (Years)	4-5	4-5	10-12	10-12	12-15	12-15	15-20	20-25
Anti-static	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Joint Life	High	High	High	Medium	High	Medium	Medium	High
Glare (Reflection)	No	Yes	No	No	No	No	Yes	No
Moisture Resistance	Low	Medium	Medium	Medium	High	High	High	High
Wear Resistance	Low	Low	Medium	Medium	High	High	High	High
Tear Resistance	High	High	Medium	Medium	High	High	High	High
Rusting	No	No	No	No	Yes	No	No	No
Air Bubble/Imprint	No	No	Yes	No	No	No	No	No
Crack Resistance	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Scratch Resistance	No	No	No	No	No	No	No	No
Cost	V. Low	Low	Low	Medium	High	High	High	V. High

EPOXY FLOORING

MERITS

- ⚡ Resistant to moisture and abrasion
- ⚡ Cost-effective

DEMERITS

- ⚡ Life of 4-5 years
- ⚡ Glare
- ⚡ Not anti-static
- ⚡ Chances of crack
- ⚡ ESD Property not available

VINYL FLOORING

SPECIFICATIONS

Completely smooth and undulated surface is must. Joints are sealed by welding rod.

MERITS

- ⚡ Cost-effective
- ⚡ Life of 10-12 years
- ⚡ Can be re-cycled
- ⚡ Glare Free

DEMERITS

- ⚡ Chances of air bubbles
- ⚡ Moisture & Abrasion Resistant - Medium

MARBLE FLOORING

MERITS

- ⚡ Resistant to moisture and abrasion
- ⚡ Life of 12-15 years
- ⚡ Glare free

DEMERITS

- ⚡ Not Jointless and seamless.
- ⚡ Not anti-static
- ⚡ ESD Property not available

FLOORING MATERIAL COMPARISON

Features	Epoxy Flooring	Vinyl Flooring	Marble Flooring
Anti-Bacterial	Yes	Yes	Yes
Anti-Fungal	Yes	Yes	Yes
Jointless & Seamless	Yes	Yes	No
Non-Porous	Yes	Yes	Yes
Washable	Yes	Yes	Yes
Thickness	65-70μ	1mm	25-30mm
Life (Years)	4-5	10-12	10-12
Anti-static	No	Yes	No
Joint Life	High	High	High
Glare (Reflection)	Yes	No	Yes
Moisture Resistance	Medium	Medium	High
Wear Resistance	Low	Medium	High
Tear Resistance	High	Medium	High
Rusting	No	No	No
Air Bubble/Imprint	No	Yes	No
Crack Resistance	No	Yes	Yes
Scratch Resistance	No	No	No
Cost	Low	Medium	Low



PPGI PUF PANEL MODULAR O.T.



HPL PUF PANEL MODULAR O.T.



SS PUF PANEL MODULAR O.T.

CLEAN ROOM DOORS

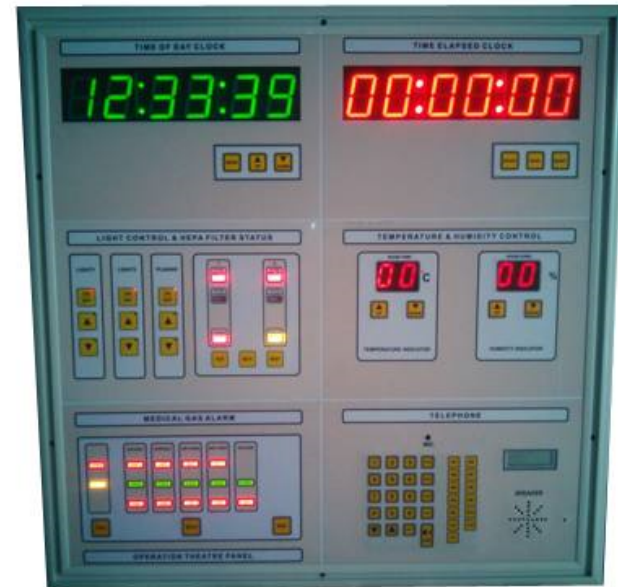
- ⚡ Sliding & Hinged Doors
- ⚡ Maintains Positive Pressure
- ⚡ Disinfecting quicker & safer
- ⚡ Acoustic Control
- ⚡ Contamination risks under control



O.T. CONTROL PANEL

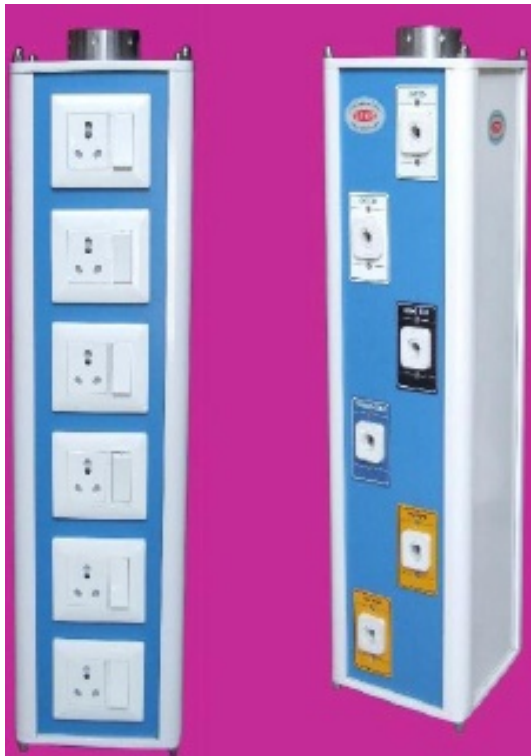
Surgeon Control Panel consisting of:

- Time of Days Clock
- Elapsed Time Clock
- Temperature Controller
- Humidity Controller
- Lighting Control/ Dimming
- Medical Gas Alarm Systems
- Hands Free Telephone
- HEPA Filter Status Indicator
- Laminar Flow Status
- OT Pressure Indicator
- Music Control



PENDANTS

- ⚡ Well designed working environment
- ⚡ Advance surgical tools/equipments
- ⚡ Positioning of supply system



HATCH BOX

- ⚡ For transfer of material
- ⚡ To minimize the entry of contaminants



PRESSURE RELIEF DAMPER

- ⚡ To Control Room Pressure



PERIPHERAL LIGHT

- ⚡ Glare Free
- ⚡ 5,000 hours life
- ⚡ CRI >90%
- ⚡ Efficiency >85%
- ⚡ Color Temperature – 6500 K
- ⚡ Max. Surface Luminance > 13,000 Lux



LED X-RAY VIEW BOX

- ⚡ 85% energy saving
- ⚡ 4-5 times long life
- ⚡ Film activated switch & auto-off
- ⚡ 12 step digital timer



STORAGE CABINET



SCRUB SINK



WRITING BOARD



VIEW WINDOW



CLIENT LIST – MODULAR O.T./ICU/IVF & ISOLATION ROOM

- ❖ Smt. Maroliya Hospital, Navsari
- ❖ Coral Hospital, Vadodara
- ❖ ASG Eye Hospital, Surat
- ❖ Metas Adventist Mission Hospital, Surat (4 OTs)
- ❖ Tristar Hospital, Surat (3 OTs)
- ❖ Dr. Prakash Modi, Visnagar
- ❖ Kothari Multi-speciality Hospital, Vadodara
- ❖ Dr. Bhargav Patel, Ahmedabad
- ❖ Surmount Hospital, Gandhinagar
- ❖ Karmdeep Hospital, Ahmedabad
- ❖ Lions Eye Hospital, Viramgam
- ❖ Shiv Hospital, Viramgam
- ❖ Care Multi-speciality Hospital, Vadodara
- ❖ Shreeji Hospital, Bodeli
- ❖ Suchak Hospital, Bhavnagar
- ❖ Apex Dialysis Center, Palanpur (OT & ICU)
- ❖ Svayambhu Hospital, Ahmedabad
- ❖ Prathana Hospital, Surat
- ❖ Deep Eye Hospital, Surat
- ❖ Gujarat Gastro & Vascular Hospital, Surat
- ❖ Satyam Hospital, Modasa
- ❖ Shiv Jyoti Hospital, Surat
- ❖ Sunshine Global Hospital, Vadodara (2 OTs)
- ❖ Sevashram Hospital, Bharuch
- ❖ Samata Hospital, Ahmedabad
- ❖ Crystal Hospital, Surat
- ❖ Silver Hospital, Surat
- ❖ Divine Medicure, Surat
- ❖ Maruti Hospital, Surat
- ❖ Global Hospital, Surat
- ❖ Gujarat Gastro & Vascular Hospital, Surat
- ❖ Sunshine Global Hospital, Vadodara (2 OTs)
- ❖ Al-Mahmood Hospital, Jambusar
- ❖ Aashvi ENT Hospital, Ahmedabad
- ❖ Anand Hospital, Ahmedabad
- ❖ Dr. Parthiv Patel, Ahmedabad
- ❖ Dr. Path Patel, Mehsana (2 OTs)
- ❖ Jeevandeep Hospital, Vapi
- ❖ Sargam Hospital, Ankleshwar
- ❖ Lions Club Hospital, Viramgam
- ❖ Lallubhai Seth Hospital, Savarkundla
- ❖ Nice Children Hospital, Viramgam (NICU)
- ❖ Vibrant Hospital, Vapi (NICU)
- ❖ Priya Nursing Home, Anand (IVF)
- ❖ Radha Hospital & Candora IVF, Surat (IVF)
- ❖ Bhagyoday Hospital, Kadi (Negative Pressure Isolation Room)
- ❖ Iris Hospital, Anand (Isolation ICU)

THANKS !!!



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